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1 Optimizers:

2 what does batch size do

3 Differences between l1 and l2 loss

The l1 normalization i.e for question about minimality, more eagerly pushes certain features or subcomponents to exact zero.

This comes as a consequence of the derivative of it.

$$\frac{d}{dx}x = 1/ - x = -1$$

$$\frac{d}{dx}x^2 = 2x$$

i.e the derivative depends on the magnitude, for l2 loss. It doesn't for l1 loss. Taken with an oscillating loss function it will oscillate to 0.

3.1 regularization

1. We don't use dropout, as this is done by the ablation anyway.
2. We don't use any weight decay, as component magnitude is quite important! Don't want feature shrinkage

4 If they ask about why a circle can't be rank 1?

5 Why does the cosine similarity go from 0 -1 ?

Cauchy schwarz inequality states at $uv \leq |u| \cdot |v|$

6 Optimizers

6.1 SGD:

$$w_i = W_{i-1} - lr * \nabla W_{i-1}$$

Issue, take p^2 , the gradient very quickly deteriorates, and goes to near zero within few iterations.

6.2 Momentum

Momentum gives memory:

$$vel = 0.5 * velocity + gradient$$

where velocity is saved over iterations. i.e it slows down slower

6.2.1 Adam:

Momentum with adagrad i.e there is a per component momentum computation, instead of a global one! So if one is more sensitive than the other its ok!

7 Quantum

Quantum (plural quanta): An indivisible computational module that, for example, retrieves a fact, implements an algorithm, or more generally corresponds to some basic skill possessed by a model.

8 example of matrix mult routing

$$W_l = \begin{bmatrix} a & b \end{bmatrix}, W_{l+1} = \begin{bmatrix} c \\ d \end{bmatrix}$$

Multiplying through gives:

$$W_{l_1} W_{l_0} = \begin{bmatrix} ac & bc \\ ad & bd \end{bmatrix}$$

Summing over these entries gives $ac + bc + ad + bd = (c + d)(a + b)$ And through the routing matrix:

$$(a + b) * (c + d)$$